



JT-DMF CRM Evaluation

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 - Benchmarking the compute node
 - Build the host capabilities file
- Media function
 - Profile the media function
 - Build the resource manifest file
- Using Kubernetes
 - Benchmark data to Resource Slice
 - DRA Consummable capacity
 - Write down a simple DRA driver
 - Media function resource manifest to DRA resource claims

Host capabilities

Benchmark

- List all the relevant compute resources available
- Multiple benchmark application
 - Easier to support multiple hardware vendors
- Example:
 - Processing speed
 - Memory capacity
 - Memory throughput
 - GPU performance
 - Network interface speed
 - Ect.

Benchmark



CPU capacity

- Number of numa node
- Number of cores
- Base frequency
- Ect.

```
root@spice02 ~/m/Linux# lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          52 bits physical, 57 bits virtual
Byte Order:             Little Endian
CPU(s):                 64
On-line CPU(s) list:   0-63
Vendor ID:              GenuineIntel
BIOS Vendor ID:         Intel(R) Corporation
Model name:             INTEL(R) XEON(R) GOLD 6526Y
  BIOS Model name:      INTEL(R) XEON(R) GOLD 6526Y   CPU @ 2.8GHz
  BIOS CPU family:      179
  CPU family:           6
  Model:                207
  Thread(s) per core:   2
  Core(s) per socket:   16
  Socket(s):            2
  Stepping:             2
  CPU(s) scaling MHz:   39%
  CPU max MHz:          3900.0000
```

Benchmark



Memory capacity

- Installed memory depth
- Memory type

```
root@spice02 ~/m/Linux# lsmem
RANGE                                SIZE  STATE  REMOVABLE  BLOCK
0x0000000000000000-0x000000007fffffff  2G  online         yes    0
0x00000000100000000-0x0000000207fffffff 126G online         yes  2-64

Memory block size:          2G
Total online memory:       128G
Total offline memory:      0B
```

Benchmark



Memory throughput

- Intel MLC
 - Support both Intel and AMD cpu
 - Support multi-cpu (NUMA)
 - [Intel MLC website](#)

```
Intel(R) Memory Latency Checker - v3.12
Measuring idle latencies for sequential access (in ns)...

           Numa node
Numa node  0      1
           0      90.1 156.1
           1     166.7 88.2

Measuring Peak Injection Memory Bandwidths for the system
Bandwidths are in MB/sec (1 MB/sec = 1,000,000 Bytes/sec)
Using all the threads from each core if Hyper-threading is enabled
Using traffic with the following read-write ratios
ALL Reads      :    114544.8
3:1 Reads-Writes :    4704.1
2:1 Reads-Writes :    1935.8
1:1 Reads-Writes :   109513.7
Stream-triad like:    774.9
All NT writes   :    89607.3
1:1 Read-NT write:    1229.2

Measuring Memory Bandwidths between nodes within system
Bandwidths are in MB/sec (1 MB/sec = 1,000,000 Bytes/sec)
Using all the threads from each core if Hyper-threading is enabled
Using Read-only traffic type

           Numa node
Numa node  0      1
           0    121423.4 113110.3
           1    112339.7 122425.7
```

Host Capabilities format:

- YAML section describing the reference platform capabilities
- All measures in the system capabilities can match a value in the requirement

Host Capabilities - CPU



```
host_capabilities:
  cpu:
    numa_nodes: 2
    number_of_cores: 16
    base_frequency: "2.8GHz"
    generation: "Emerald Rapids" # CPU microarchitecture generation (e.g., Skylake, Ice Lake, etc.)
    # Selected CPU features discovered via NFD or cpuid
    features:
      - AVX512F
      - AVX512DQ
      - SGX
```

Host Capabilities - Memory



```
memory:
  - node: 0
    type: ram
    generation: DDR5
    size: 128Gi
    throughput:
      total: 122Gi
  - node: 1
    type: ram
    generation: DDR5
    size: 128Gi
    throughput:
      total: 121Gi
```

Media function profiling

Profiling :

- Measures the resources requirement to run a specific media function
 - CPU usage
 - Memory Usage
 - Memory throughput
 - Ect.
- All measured values must match a capacity available in the host capabilities

Media Function profiling



CPU requirements:

- Measure CPU usage on the reference platform
- Uses perf stat tool
- Difficult to extrapolate the usage when changing CPU type

```
spice02@spice02:~$ sudo perf stat -p 215504

Performance counter stats for process id '215504':

      7,128.46 msec task-clock                #    0.575 CPUs utilized
         4,834      context-switches         #   678.127 /sec
          1,342      cpu-migrations          #   188.259 /sec
        168,865      page-faults            #    23.689 K/sec
 26,523,553,701      cycles                  #    3.721 GHz
 96,556,334,522      instructions            #    3.64 insn per cycle
  7,906,506,401      branches                #    1.109 G/sec
   15,925,823      branch-misses            #    0.20% of all branches

12.405488652 seconds time elapsed
```

Media Function Profiling



Memory Usage

- Sample the process PID status
- Compute memory usage using:
 - RssAnon
 - RssFile
 - RssShmem

```
spice02@spice02:~$ sudo cat /proc/325813/status
Name:      mxl-gst-sink
Umask:     0002
State:      S (sleeping)
Tgid:      325813
Ngid:      0
Pid:       325813
PPid:      214969
TracerPid: 0
Uid:       1000    1000    1000    1000
Gid:       1000    1000    1000    1000
FDSize:    256
Groups:    4 24 27 30 46 101 987 1000
NSTgid:    325813
NSpid:     325813
NSpgid:    325813
NSsid:     214969
Kthread:   0
VmPeak:    1356804 kB
VmSize:    919680 kB
VmLck:     0 kB
VmPin:     0 kB
VmHWM:     519108 kB
VmRSS:     122156 kB
RssAnon:   55148 kB
RssFile:   12904 kB
RssShmem:  54104 kB
VmData:    244576 kB
VmStk:     132 kB
VmExe:     724 kB
VmLib:     11228 kB
VmPTE:     540 kB
VmSwap:    0 kB
HugetlbPages: 0 kB
```

Memory Throughput

- Not available per PID, only system wide
- Compute by:
 - Setup all the MXL producer
 - Measure the system memory throughput
 - Add the UUT (unit under test)
 - Measure again
 - Memory throughput $\approx$ After - before

Media Function Profiling



Memory Throughput

- Proprietary software
 - Intel: pcm-memory
 - AMD: AMDuprofPerf
- Uses low level CPU register
- Need to run on bare metal
 - Any other tools available?
 - What about cloud?

Before

```
Detected INTEL(R) XEON(R) GOLD 6526Y "Intel(r) microarchitecture codename Emerald Rapids-SP"
Update every 1 seconds
Read/Write values expressed in (MB/s)
Date,Time,Read,Write,Memory
2026-04-28,13:43:42.931,2388.41,1548.77,3937.18
2026-04-28,13:43:43.931,3296.94,1343.39,4640.34
2026-04-28,13:43:44.932,1766.84,1056.34,2823.18
2026-04-28,13:43:45.932,1661.70,933.10,2594.79
2026-04-28,13:43:46.932,1738.46,896.93,2635.39
2026-04-28,13:43:47.932,1685.27,890.55,2575.83
2026-04-28,13:43:48.932,1643.17,897.10,2540.27
2026-04-28,13:43:49.932,1676.44,982.63,2659.07
2026-04-28,13:43:50.932,1777.67,1161.28,2938.95
2026-04-28,13:43:51.932,1645.46,1078.77,2724.23
Cleaning up
Closed perf event handles
Zeroed uncore PMU registers
Freeing up all RMIDs
```

After

```
Detected INTEL(R) XEON(R) GOLD 6526Y "Intel(r) microarchitecture codename Emerald Rapids-SP"
Update every 1 seconds
Read/Write values expressed in (MB/s)
Date,Time,Read,Write,Memory
2026-04-28,13:42:39.651,3086.06,2439.40,5525.46
2026-04-28,13:42:40.651,2866.04,2102.64,4968.68
2026-04-28,13:42:41.651,2909.53,2101.16,5010.69
2026-04-28,13:42:42.651,3979.31,2928.77,6908.07
2026-04-28,13:42:43.651,4919.51,3187.74,8107.26
2026-04-28,13:42:44.651,3136.21,2381.27,5517.49
2026-04-28,13:42:45.651,3130.33,2381.53,5511.86
2026-04-28,13:42:46.651,3116.27,2264.06,5380.34
2026-04-28,13:42:47.651,3111.51,2311.79,5423.30
2026-04-28,13:42:48.651,2994.69,2149.64,5144.33
Cleaning up
Closed perf event handles
Zeroed uncore PMU registers
Freeing up all RMIDs
```

Media Function Profiling



YAML media function definition

- Include all the media function arguments
- Uses to define the resource requirements

```
apiVersion: mxl.media.fn/v1alpha1
kind: MxlMediaFunctionParameters
metadata:
  name: mxl-gst-sink-1080p30-v210
  label:
    app: mxl-gst-sink
    function: video-sink
    args:
      - # All arguments that are relevant
        resolution: "1080p"
        frame_rate: "30fps"
        format: "v210"
    namespace: media
spec:
  role: video-sink
  flow:
    label: "multiviewer-1"

  # Media input specifications
  inputs:
    feed-1:
      format:
        video:
          media_type: video/v210
          frame_width: 1920
          frame_height: 1080
          frame_rate: "30000/1001"
    feed-2:
      format:
        video:
          media_type: video/v210
          frame_width: 1920
          frame_height: 1080
          frame_rate: "30000/1001"

  # Media output specifications
  outputs:
    processed-output:
      format:
        video:
          media_type: video/v210
          frame_width: 1280
          frame_height: 480
          frame_rate: "30000/1001"
```

Media Function Profiling



Media function resources requirements

- Match the host capability structure
- List only the needed requirements

```
# Resource requirements for the media function
# Similar to the host capabilities, the value in the requirement match the unit used in the reference platform,
# but here it represents the actual resource request/limit for this media function.
# Uses Kubernetes-like resource specification for consistency, but can be extended with custom fields as needed.
requirements:
  - cpu:
      - cpu: "600m"
      - cpu_class: "performance"
      - features:
          - null

  - memory:
      - type: ram
      - size: "1Gi" # Requests and limits in units as expressed in Kubernetes (e.g., Mi, Gi)
      - throughput:
          - read: "1500MBi"
          - write: "1500MBi"
      - total: "3Gi" # Total memory bandwidth required (if applicable)
```

Kubernetes example

Natively supported resources

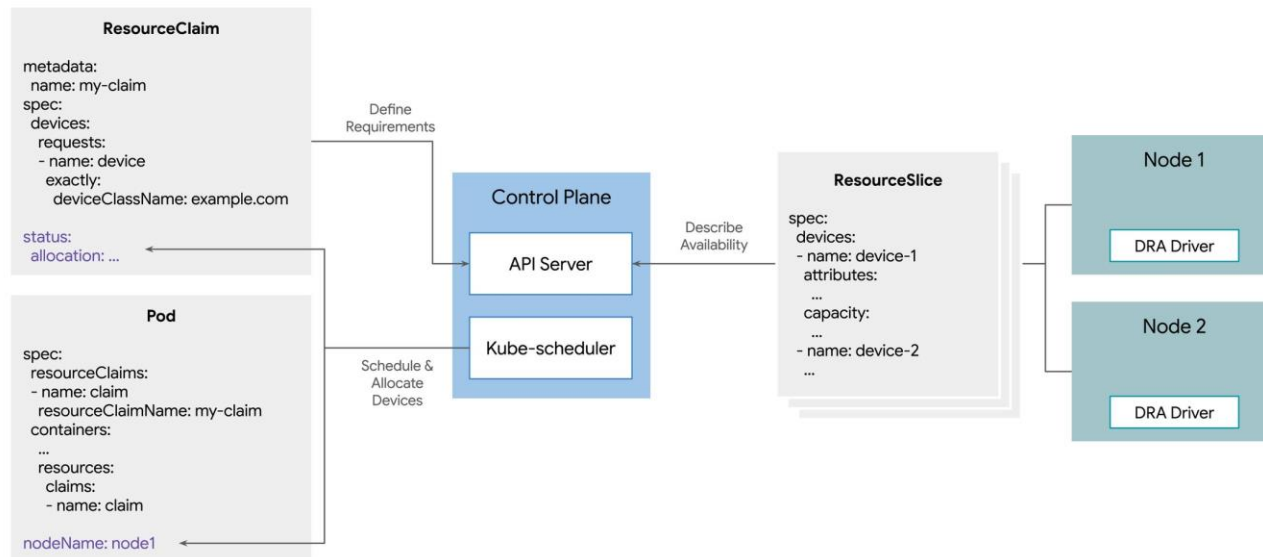
- CPU
- Memory
- Ephemeral-storage
- Hugepages Size

DRA : Dynamic Resource Allocation (GA Kubernetes 1.35)

- Any resources can be added through DRA
- Can manage both virtual and real resources
- Resource allocation is managed by the Kubernetes scheduler
- Leverage Consumable capacity and Multiple allocations
 - Feature still in beta (Kubernetes 1.36)

DRA Driver

- Custom software for each resources
- Run on each node
- Publish the node resource slice for its resource class



Memory Throughput DRA



Definition of the resource slice

- Custom DRA driver for each resources
- Publish the available memory throughput for each numa node.
- <https://github.com/kubernetes-sigs/dra-example-driver>

```
apiVersion: v1
items:
- apiVersion: resource.k8s.io/v1
  kind: ResourceSlice
  metadata:
    creationTimestamp: "2026-05-15T10:38:06Z"
    generateName: 00000-mem.example.com-dra-memory-driver-cluster-worker-
    generation: 1
    name: 00000-mem.example.com-dra-memory-driver-cluster-worker-jgchw
    ownerReferences:
      - apiVersion: v1
        controller: true
        kind: Node
        name: dra-memory-driver-cluster-worker
        uid: 6271d5f1-ce8e-4bf0-ad6e-18a544f110f9
    resourceVersion: "38388"
    uid: 30abf292-f0ad-4361-b8e7-839864983811
  spec:
    devices:
      - allowMultipleAllocations: true
        attributes:
          numa:
            int: 0
          capacity:
            mem:
              value: 121Gi
            name: numa-0
        allowMultipleAllocations: true
        attributes:
          numa:
            int: 1
          capacity:
            mem:
              value: 122Gi
            name: numa-1
        driver: mem.example.com
        nodeName: dra-memory-driver-cluster-worker
        pool:
          generation: 1
          name: dra-memory-driver-cluster-worker
          resourceSliceCount: 1
    kind: List
    metadata:
      resourceVersion: ""
```

Kubernetes pod



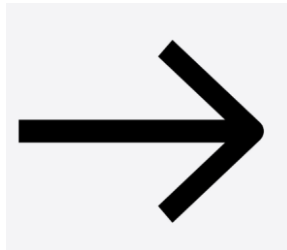
Translate Media function resources requirements to K8s resources claims

- Requires 0.6 CPU
- 1GB of memory
- 3GB/s of memory throughput

CRM Resources requirements

```
requirements:
  cpu:
    - cpu: "600m"
    - cpu_class: "performance"
    - features:
        - null

  memory:
    - type: ram
    - size: "1Gi"
    - throughput:
        read: "1500MBi"
        write: "1500MBi"
        total: "3Gi"
```



Kubernetes resources claims

```
apiVersion: resource.k8s.io/v1
kind: ResourceClaimTemplate
metadata:
  namespace: media
  name: mxl-gst-sink-1080p30-v210-memory-bandwidth-claim-template
spec:
  spec:
    devices:
      requests:
        - name: mem
          exactly:
            deviceClassName: mem.example.com
            capacity:
              requests:
                mem: "3Gi"
  resources:
    limits:
      cpu: "1"
      memory: "1Gi"
    requests:
      cpu: "600m"
      memory: "1Gi"
  resourceClaims:
    - name: mxl-gst-sink-1080p30-v210-memory-bandwidth-claim
      resourceClaimTemplateName: mxl-gst-sink-1080p30-v210-memory-bandwidth-claim-template
```

Conclusion

Conclusion



Questions?